

9) Pin Description of MN6617

● IC301: MN6617 (Digital Signal Processing: EFM Decoder, Error Correction, CLV Servo)

Pin No.	Mark	I/O Division	Function
1	BLKCK	O	Sub-code block (Q data) clock (75Hz)
2	CLDCK	O	Sub-code frame (Q data) clock (7.35kHz)
3	SUBQ	O	Sub-code (Q data) output
4	CRC	O	Sub-code (Q data) CRC check (Not used, open)
5	RST	I	Reset signal input (reset at "L")
6	MLD	I	Command load input
7	MCLK	I	Command clock input
8	MDATA	I	Command data input
9	DMUTE	I	Muting control (Not used, connected to GND)
10	TRON	I	Tracking servo ON signal (tracking servo ON at "L")
11	STAT	O	Processing condition (CRC, OTC, CLVOK, TT STOP) output
12	SMCK	O	Clock output (4.2336MHz)
13	PMCK	O	Pitch control clock output (Not used, open)
14	ITC	I	Track counter input signal (Not used, open)
15	TEST	I	Test mode selection (Not used, connected to +5V)
16	X2	O	Clock output (Not used, open)
17	X1	I	Clock input
18	SEL	I	DA output parallel/serial selection (serial at "L") (Not used, connected to GND)
19	LDG/WDC	O	L channel deglitch signal/serial data word clock. (Not used, open)
20	RDG	O	Clock signal output for spindle motor control (Not used, open)
21	DEMPH	O	De-emphasis ON signal (de-emphasis ON at "H") (Not used, open)
22	IPFLAG	O	Interpolation flag (interpolation at "H") (Not used, open)
23	FLAGO	O	Error flag (error at "H") (Not used, open)
24	FLAG6	O	16K RAM address reset signal (reset at "H") (Not used, open)
25	XCK	O	Clock (16.9344MHz) output (Not used, open)
26	SRDATA	O	Serial data output (MSB first)
27	SRDATA	O	Serial data output (LSB first) (Not used, open)
28	SRCK	O	Serial data beat clock.
29	WDCK	O	Serial data word clock (Not used)
30	BYTCK	O	Serial data byte clock (Not used, open)
31	GND	I	GND terminal
32	R/L	O	R/L signal

Pin No.	Mark	I/O Division	Function
33	RESY	O	Resynchronizing signal (Not used, open)
34	FCLV	O	Synchronizing detection signal (Not used, open)
35	IPBYTE	O	Interpolation flag for each byte (Not used, open)
36	IPSEL	I/O	Interpolation prohibited (Not used, connected to GND)
37	FLAG5	O	C2 decoder correction flag 3 (Not used, open)
38	FLAG4	O	C2 decoder correction flag 2 (Not used, open)
39	FLAG3	O	C2 decoder correction flag 1 (Not used, open)
40	FLAG2	O	C1 decoder correction flag 2 (Not used, open)
41	FLAG1	O	C1 decoder correction flag 1 (Not used, open)
42	FLCK	O	Crystal frame clock
43 50	D7 D0	I/O	16K RAM data output
51	RAMOE	O	16K RAM $\overline{OE}$ signal
52	RAMWE	O	16K RAM $\overline{WE}$ signal
53 63	RAMA0 RAMA10	O	16K RAM address signal (RAMA0: LSB, RAMA10: MSB)
64	PC	O	Spindle motor ON signal (ON at "L")
65	EC	O	Spindle motor drive signal
66	FG	I	Spindle motor FG signal input (Not used, open)
67	VCNT	—	(Not used, open)
68	REXT	—	(Not used, open)
69	VDD	I	Power supply (+5V)
70	PD	—	(Not used, open)
71	PCKO	—	(Not used, open)
72	PCK	I	PLL extract clock input
73	VDD	I	Power supply (connected to +5V)
74	EFM	I	EFM signal input (PLL)
75	SRF	I	EFM signal input (DSL)
76	DO	I	Drop-out signal (Drop-out at "H")
77	CLVS	O	11T servo OK signal (OK at "H") (Not used, open)
78	FPC	O	PLL frequency comparison signal
79	BSSEL	O	PLL frequency in take operation signal. (Not used, open)
80	SRFO	—	(Not used, open)
81	NSRF	—	(Not used, open)
82	RF	—	(Not used, open)
83	SUBC	O	Sub-code serial output data (Not used, open)
84	SBCK	I	Clock for sub-code serial output (Not used, open)

Table 18-5

10) Block Diagram of Digital Circuit

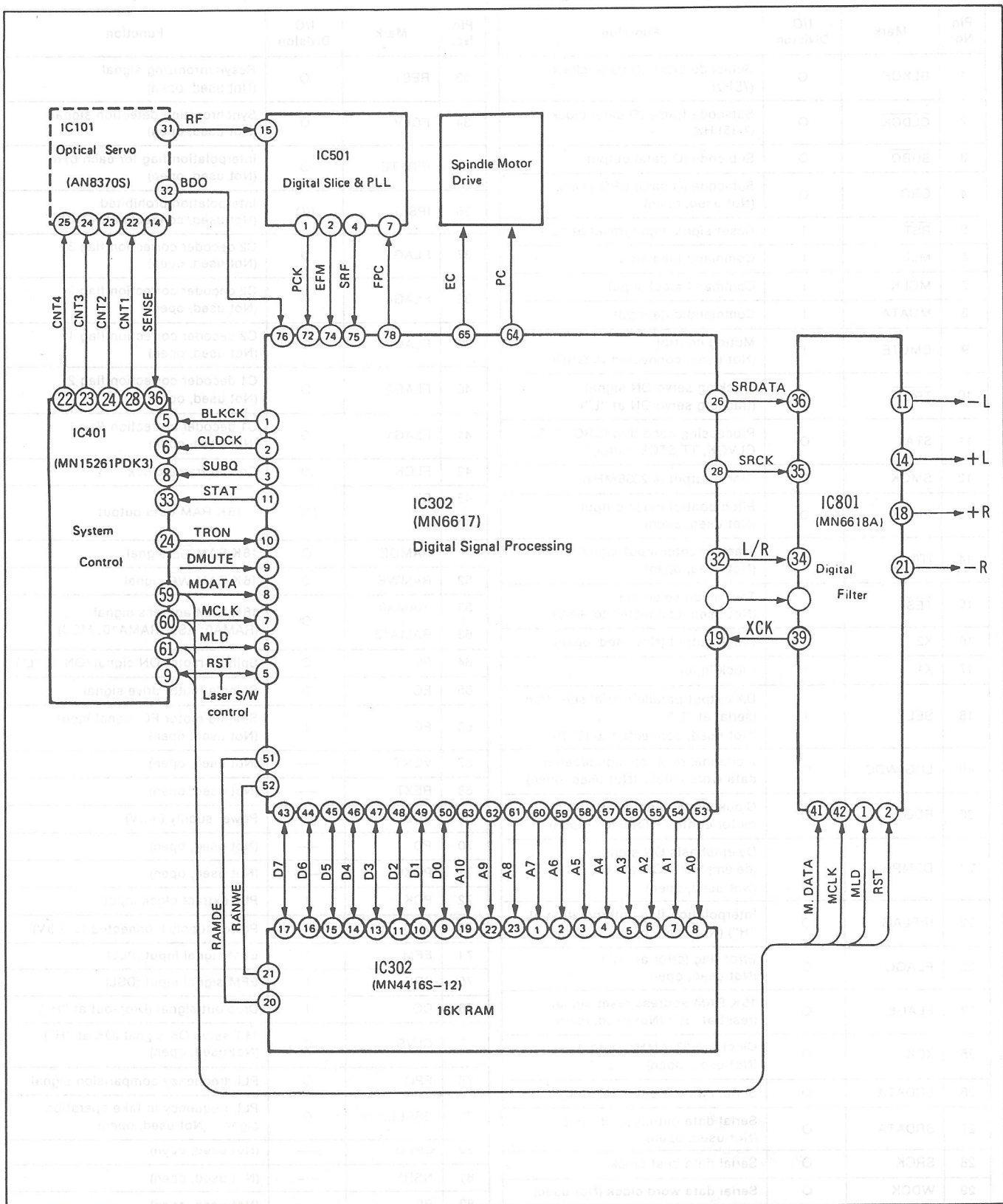


Figure 18-16

3) The Effect of the Sampling Frequency

Another important factor is sampling frequency, for efficiency in digital operations, this is kept low (44.1KHz) at a little over twice the maximum audio frequency (20.0KHz). During the sampling process this frequency and its harmonics or multiples become modulated with the original waveform in a series of frequency bands as shows figure 19-2.

The bottom end of the lowest band is rather close to the top of the audio frequency range.

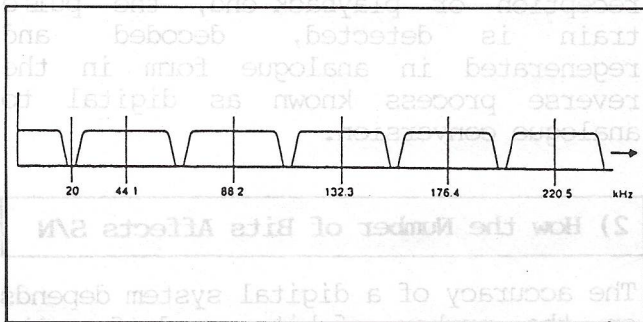


Figure 19-2

Figure 19-2 shows the frequency spectrum produced by sampling showing the periodic repetitions of the audio band. The bottom end of the lowest band is close to the audio frequency range.

4) Quantization Noise

In a digital audio system the audible noise has a different character to the familiar noise in an analogue audio system. The digital audio system background noise is called quantization noise.

The quantization noise of a system can be defined as the error signal between the original waveform before the analogue to digital conversion process, and the restored waveform after the digital to analogue conversion process. This error signal can be considered as both noise and distortion at the same time.

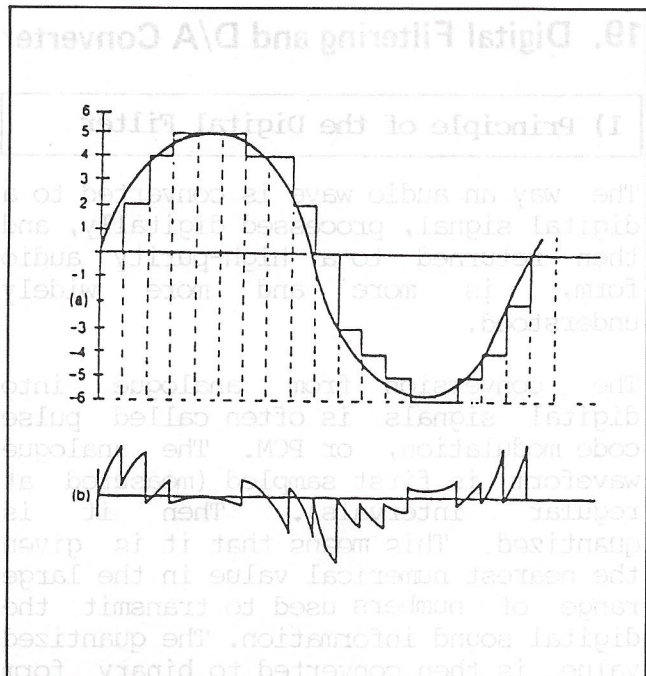


Figure 19-3

Figure 19-3 shows the quantization noise, The difference between the original audio waveform and the regenerated wave after D/A conversion (a) causes quantization noise (b).

The 16 bits analogue to digital conversion at the recording stage delivers a 16 bits digital signal with an intrinsic accuracy or signal to noise ratio of maximum 96dB.

An ideal digital to analogue converter (D/A) would deliver an analogue output signal without adding noise to it. However, ideal D/A do not exist a small but finite electronic inaccuracy adds noise to the quantization noise of the digital signal.

The inaccuracy of a 16 bits D/A (the resolution limit of half the least significant bit) can be interpreted as a noise source that reduces the signal to noise ratio of the D/A to 96dB.



19. Digital Filtering and D/A Converter

1) Principle of the Digital Filter

The way an audio wave is converted to a digital signal, processed digitally, and then returned to a high-purity audio form, is more and more widely understood.

The conversion from analogue into digital signals is often called pulse code modulation, or PCM. The analogue waveform is first sampled (measured at regular intervals). Then it is quantized. This means that it is given the nearest numerical value in the large range of numbers used to transmit the digital sound information. The quantized value is then converted to binary form and encoded into a pulse train representing the series of numbers. This process of analogue to digital conversion is shown in figure 19-1.

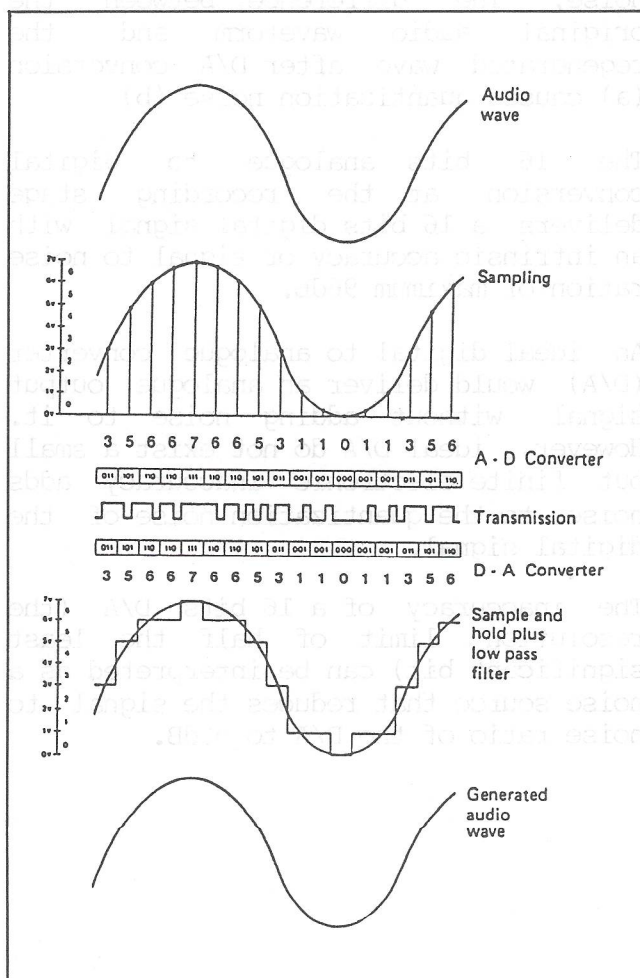


Figure 19-1

Figure 19-1 shows the digital sound reproduction process. The audio signal is sampled, quantized, converted to binary form, and encoded for recording or transmission.

Reversing the process produces a replica of the original signal. It is in this second stage that CD players use digital filtering to achieve exceptionally accurate reproduction.

The pulse train is fed into the transmission system (a cable or radio link, a tape, or a disc etc). At the reception or playback end, the pulse train is detected, decoded and regenerated in analogue form in the reverse process known as digital to analogue conversion.

2) How the Number of Bits Affects S/N

The accuracy of a digital system depends on the number of bits used for the binary number. This is because the number of bits determines the accuracy with which the samples taken from the original waveform can be quantized, i.e. expressed in digital form.

Two bits give four possible values for use in quantization, also three bits give eight possible values.

Every bit added doubles the number of possible values. In audio terms, this adds 6dB to the signal to noise ratio, because it reduces the quantization noise by a factor of 2.

The sixteen bits as used in the CD system give 65536 (2^{16}) possible values for quantization, and a signal to noise ratio of 96dB. The audio signals are encoded in the actual CD with this ratio.

96dB is an approximation. An additional factor, which depends on the type of signal, is always present, regardless of the number of bits; for sine wave signals this makes the theoretical figure 97.8dB.

The effect of 16 bits analogue to digital conversion followed by 16 bits digital to analogue conversion with 1/2 LSB accuracy is to double the noise. In consequence, the maximum signal to noise ratio achievable in practice with a 16 bit system such as CD digital audio is 93dB.

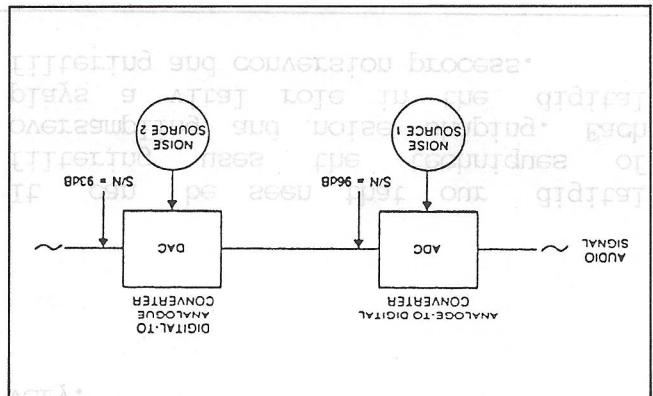


Figure 19-4

Figure 19-4 shows the signal to noise ratio deterioration in a digital chain, consisting of an A/D and a practical D/A.

5) Need for Filtering

With 16 bits (65536 possible numbers) and a sampling frequency of 44.1KHz, the CD system is theoretically capable of far higher reproduction quality than any conventional record player. But to achieve this quality, the output signal that the player derives from the disc must contain only the original audio frequencies, and not the periodic repetitions of the audio frequencies produced by the sampling (see figure 19-2).

These frequency bands can overload the power amplifier, and mix with audio frequencies or other signals to cause audible interference. They must be filtered out.

6) How D/A Conversion Operate

Figure 19-5 shows the generation of sample (a) and hold (b) produces a step waveform closer to the original.

In the pulse train a series of binary number represents quantized samples of the original audio frequency. If these are simply converted to analogue values, a series of sample is produced (see figure 19-5).

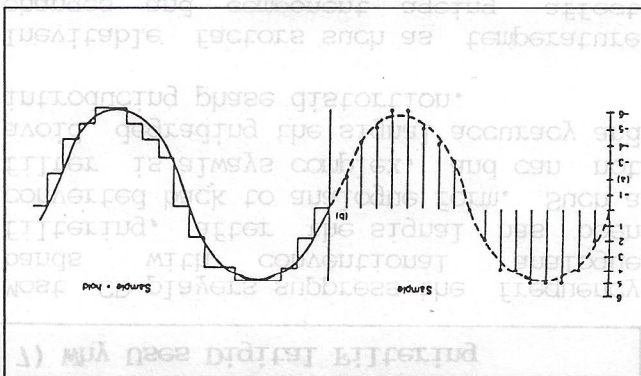


Figure 19-5

The outline of this series resembles the audio wave, but the constant on and off switching produces the infinite series of frequency bands which was shown in figure 19-2. In practice, each sample is held in the converter until the next sample arrives, to produce a step waveform which is much nearer the shape of the audio waveform than the series of sample.

This sample and hold process is illustrated in figure 19-5 (b). It substantially reduces the frequency band components, as can be seen from figure 19-6. Significant components still remain, however, particularly in the region near the audio band.

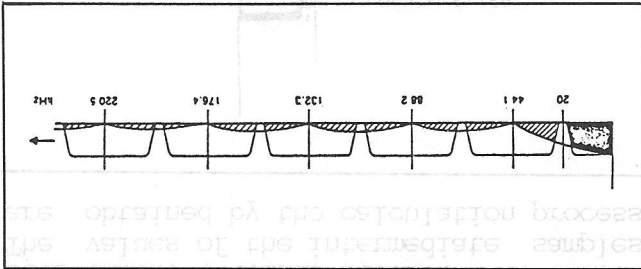


Figure 19-6

Figure 19-6 shows the effect of sample and hold. The remaining signal components are shown shaded.

7) Why Uses Digital Filtering

Most CD players suppress the frequency bands with conventional analogue filtering, after the signal has been converted back to analogue form. Such a filter is always complex, and can not avoid degrading the signal accuracy and introducing phase distortion.

Inevitable factors such as temperature changes and component ageing affect performance. Most CD players on the

8) Digital Filtering Process

A schematic diagram of the complete our D/A conversion system including digital filtering, is shows in figure 19-7.

other hand, carry out this essential filtering process digitally, with no loss of accuracy, and phase distortion. Digital filtering is a pure numerical calculation process with fixed coefficients. The fixing of the filter coefficients determines the filter characteristics which cannot afterwards very.

It can be seen that our digital filtering uses the techniques of oversampling and noise shaping. Each plays a vital role in the digital filtering and conversion process.

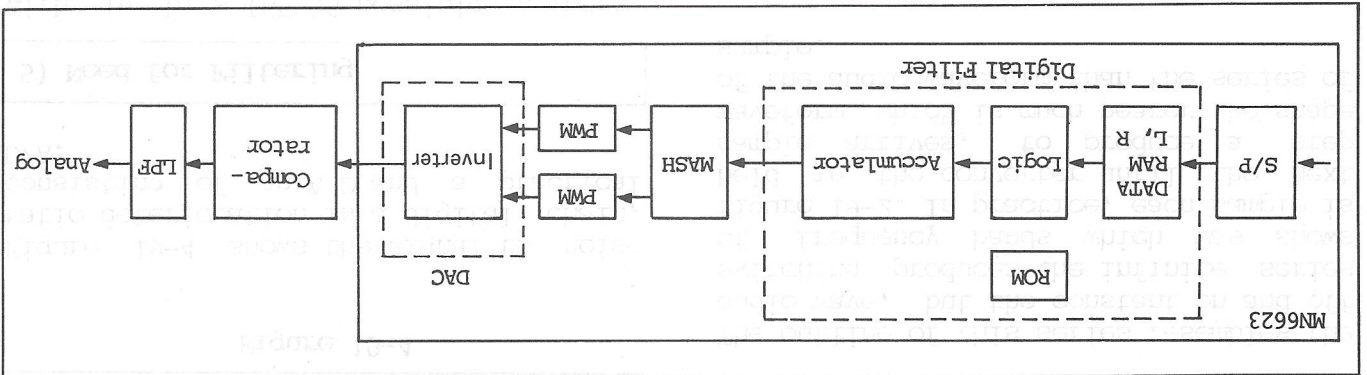


Figure 19-7

Oversampling:

Oversampling simply means multiplying the sampling frequency, in this case by four times. (44.1KHz becomes 176.4KHz.) The values of the intermediate samples are obtained by the calculation process

which determines the filter characteristic. An immediate effect of the oversampling as shows in figure 19-8 is to suppress all the lower frequency bands, centered on fs, 2fs, and 3fs.

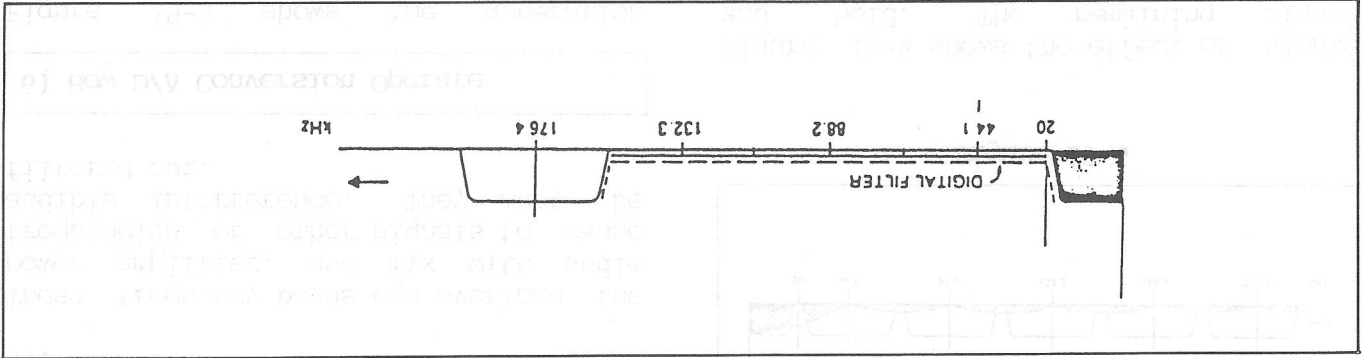


Figure 19-8

Digital Filter Block

The digital filter block consists of three sub-blocks: ATT, FIR1, and FIR2 as shown in figure 19-10. It performs digital attenuation and oversampling to convert the input 16 bits, f_s data into a 16 bits, $4f_s$ signal.

The ATT attenuates the input data in a 1/64 increment by multiplying the data by any 6 bits data. FIR1 and FIR2 are 2-fold oversampling digital filters with 85 and 21 taps each. Four-fold oversampling is accomplished by cascading the two fold oversampling filters.

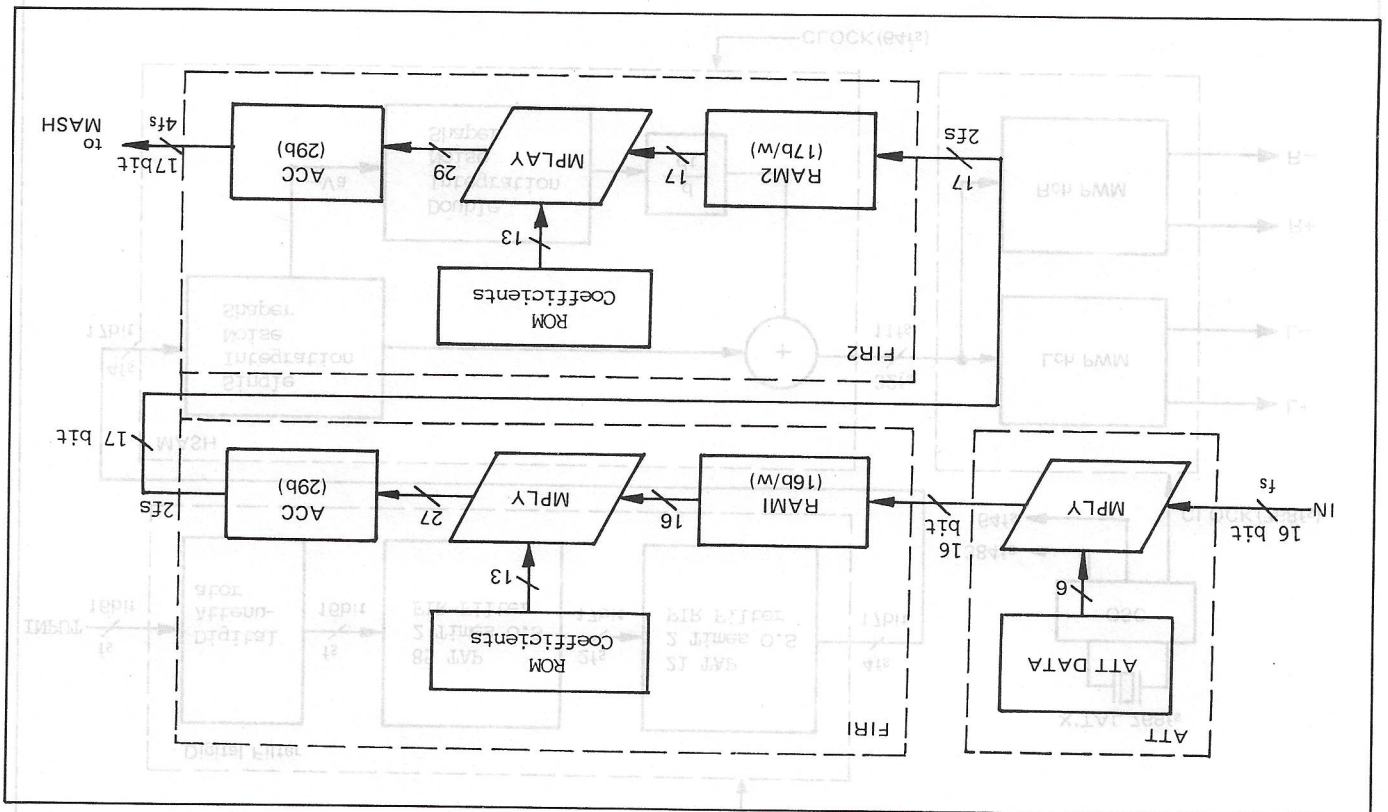


Figure 19-10 Digital Filter Block Diagram

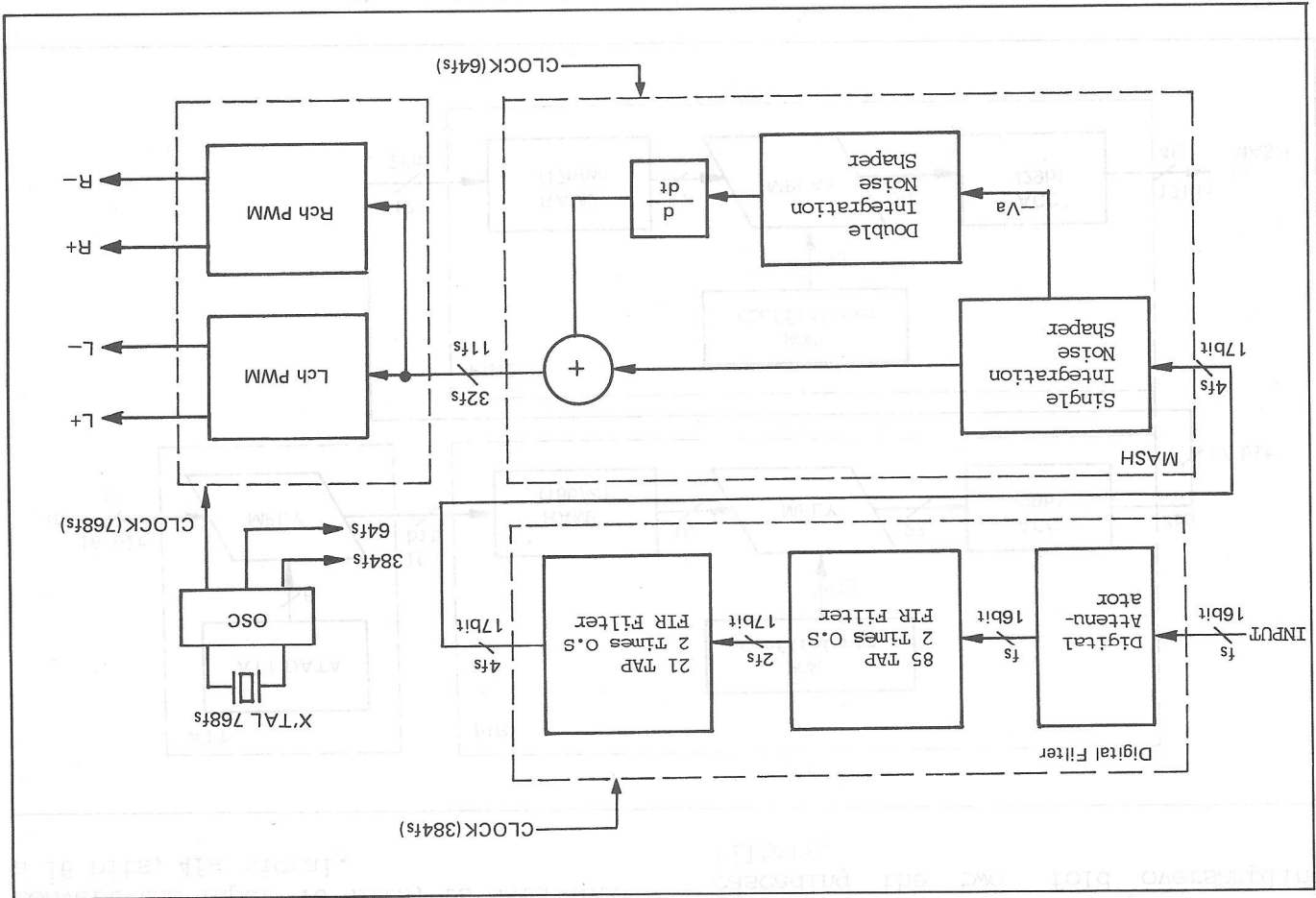
The digital filter block contains many memories (delay logics), as shown in the block diagram above, with which it computes ROM coefficients at high speed while summing the coefficients to interpolate them. For more detail, shows in figure 19-11.

where, $4f_s$ is the output signal, f_s is the input signal, and $1/64$ is the attenuation factor. The digital filter block contains many memories (delay logics), as shown in the block diagram above, with which it computes ROM coefficients at high speed while summing the coefficients to interpolate them. For more detail, shows in figure 19-11.

The digital filter and MASH blocks use 384fs and 64fs clocks, respectively, to perform time-division multiplexing on the left and right channel signals. The PWM block uses a 768fs clock to process the left and right signals independently. The MASH block converts the 16 bits, 32fs signal into 11 values, which after Pulse-Width Modulation in the PWM block become analog signals. The following section describe the function of each MN6623 block.

The MASH block operates at 32fs, where fs is the sampled input signal frequency. This means that a 32 fold oversampling filter is required. On the MN6623, however, the first digital filter oversamples the fs signal into a 4fs signal. The following MASH block then performs zero-order holding on the 4fs signal to convert it into a 32fs signal.

Figure 19-9 MN6623 Block Diagram



The MN6623 D/A converter system is composed of a digital filter, MASH, and PWM blocks as shown in figure 19-9.

9) MN6623 Actual Operation

MASH Block

The MASH block consists of many noise shapers connected in series. The noise shapers include the signal and double integration types.

* Single integration noise shaper

The single integration noise shaper uses only a single integrator for noise shaping as shows in figure 19-12. The block marked "Q" in this figure is a local quantizer which adds quantization error to the input data. The relationship between output (Y) and input (X) for this type of noise shaper is:

$$Y = X + (1 - Z^{-1}) \cdot V_q \dots\dots\dots (1)$$

The quantization error, V_q is a random value and the factor $(1 - Z^{-1})$ in this equation represents a differentiation response. Eq. (1) tells us, therefore, that the spectrum of the quantization error, V_q in the single integration noise shaper has an attenuation slope of -6 dB/oct toward lower frequencies as shows in figure 19-14.

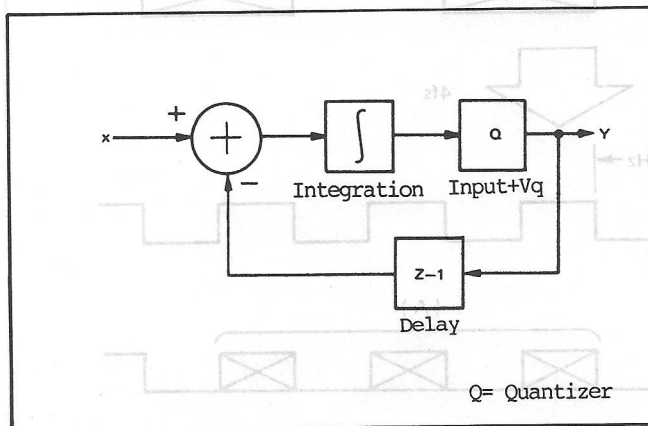


Figure 19-12 Single Integration Noise Shaper

* Double integration noise shaper

The double integration noise shaper uses two integrations in tandem as shows in figure 19-13. When the circuit is viewed at point W toward the output Y, it is

identical to the single integration noise shaper. Therefore, the relationship between Y and W is:

$$W = W + (1 - Z^{-1}) \cdot V_q \dots\dots\dots (2)$$

W expressed as a function of X is:

$$W = \frac{1}{1 - Z^{-1}} (X - Z^{-1} \cdot Y) \dots\dots\dots (3)$$

From Eqs. (2) and (3) we obtain following overall output-input relationship:

$$Y = X + (1 - Z^{-1})^2 \cdot V_q$$

If we compare this equation with Eq. (1), we find that the element $(1 - Z^{-1})$ is squared in the former equation.

This means that with the double integration noise shaper, the spectrum of the quantization error, V_q , has an attenuation slope of 12 dB/oct toward lower frequencies.

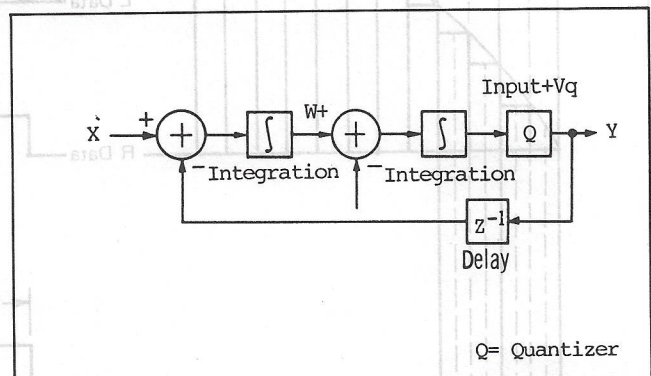


Figure 19-13 Double Integration Noise Shaper

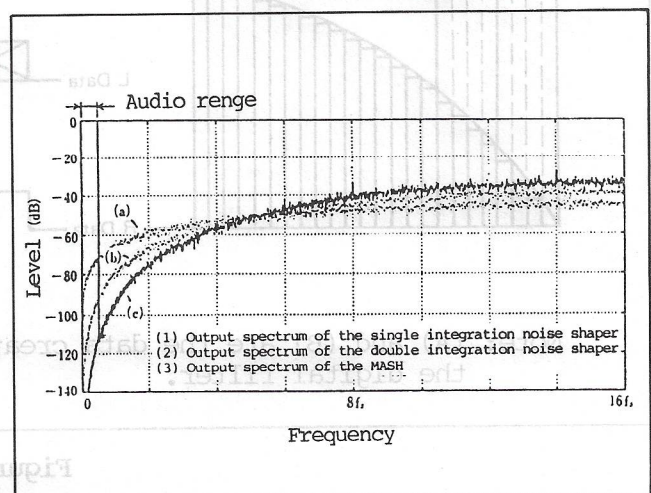
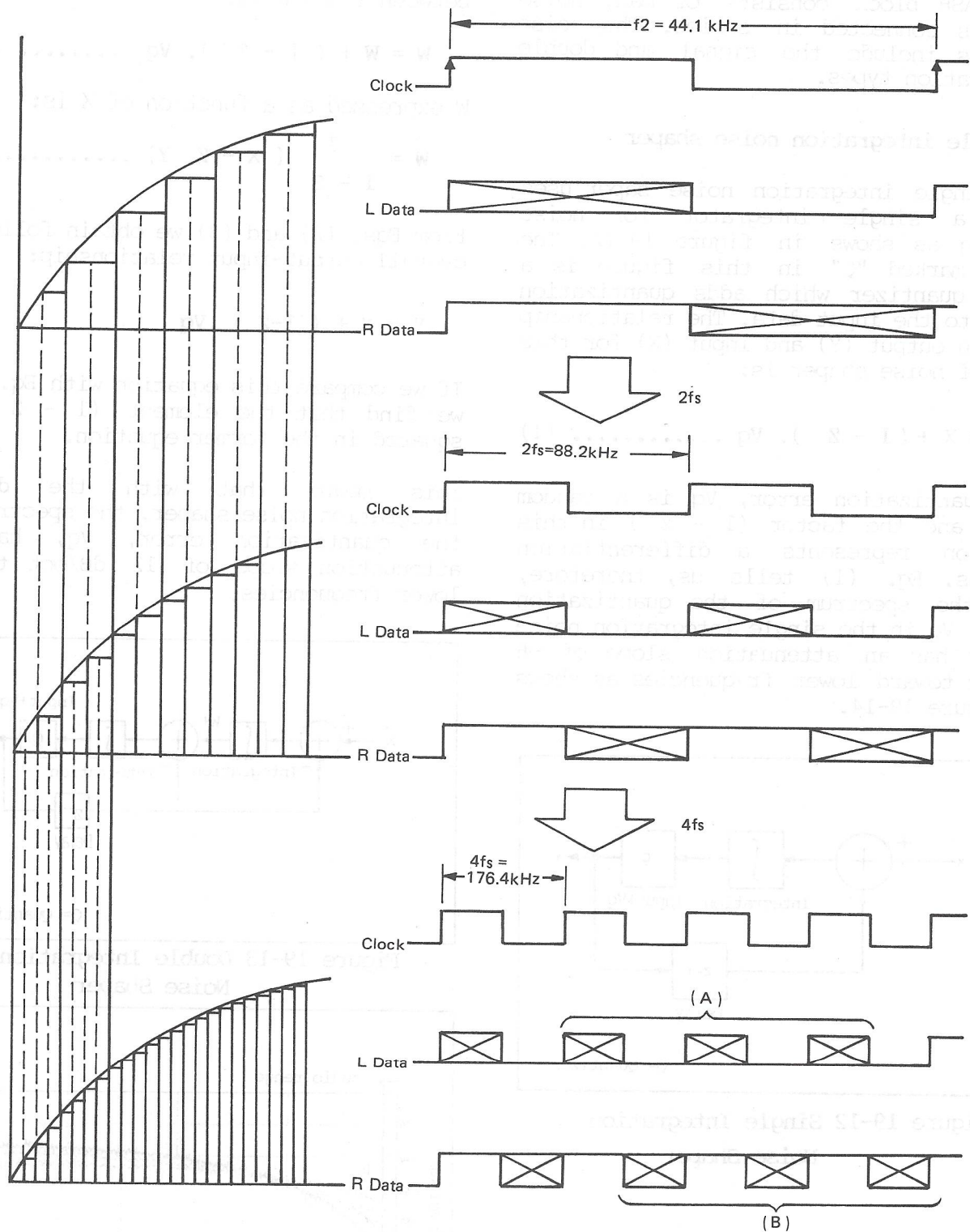


Figure 19-14 Noise Shaper Output Spectrum



Note: (A) and (B) are the data created in the digital filter.

Figure 19-11

The following chart contains the timings of the 11 types of digital data yielded by the MASH block.

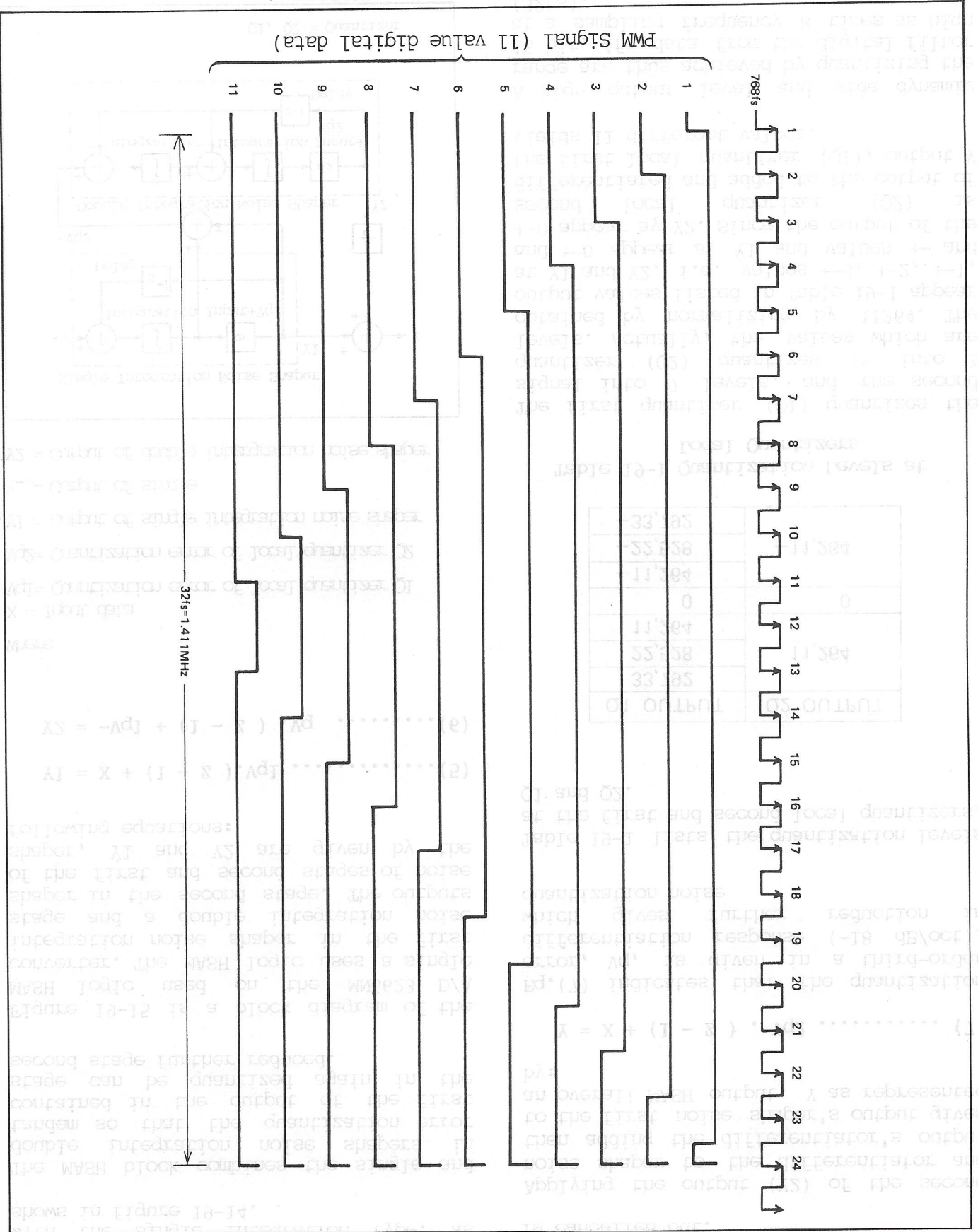


Figure 19-16 Output Data from MASH Part

The double integration noise shaper thus provides a greater attenuation of quantization noise than that obtained with the single integration type, as shows in figure 19-14.

The MASH block combines the single and double integration noise shapers in tandem so that the quantization error contained in the output of the first stage can be quantized again in the second stage further reduced.

Figure 19-15 is a block diagram of the MASH logic used on the MN6623 D/A converter. The MASH logic uses a single integration noise shaper in the first stage and a double integration noise shaper in the second stage. The outputs of the first and second stages of noise shaper, V_1 and V_2 are given by the following equations:

$$V_1 = X + (1 - Z) \cdot Vq_1 \dots \dots \dots (5)$$

$$V_2 = -Vq_1 + (1 - Z) \cdot Vq_2 \dots \dots \dots (6)$$

Where

X = Input data

Vq_1 = Quantization error of local quantizer Q_1

Vq_2 = Quantization error of local quantizer Q_2

V_1 = Output of single integration noise shaper

V_2 = Output of double integration noise shaper

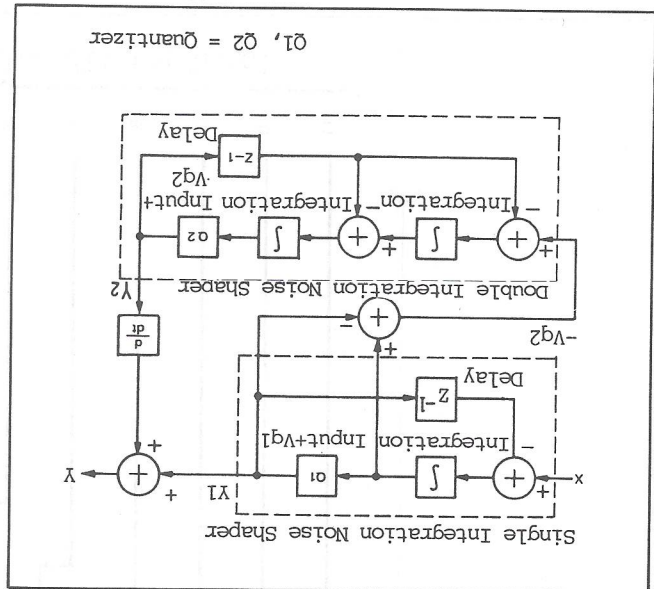


Figure 19-15 MASH Block Diagram

If we multiply both members of Eq.(5) by $(1 - Z)$ and add them to those of Eq. (6), we find that the quantization error is cancelled out.

Applying the output (V_2) of the second noise shaper to the differentiator and then adding the differentiator's output to the first noise shaper's output gives an overall MASH output, Y as represented by:

$$Y = X + (1 - Z) \cdot Vq_2 \dots \dots \dots (7)$$

Eq.(7) indicates that the quantization error, Vq , is given in a third-order differentiation response (-18 dB/oct), which gives further reduction in quantization noise

Table 19-1 lists the quantization levels at the first and second local quantizers, Q_1 and Q_2 .

Q1 OUTPUT	Q2 OUTPUT		
	33,792	22,528	11,264
11,264	22,528	11,264	0
	11,264	0	0
	0	-11,264	-22,528
-11,264	-11,264	-22,528	-33,792

Table 19-1 Quantization Levels at Local Quantizers

The first quantizer (Q_1) quantizes the signal into 7 levels, and the second quantizer (Q_2) quantizes it into 3 levels. Actually, the values which are obtained by normalizing by 11264. The output values listed in Table 19-1 appear at V_1 and V_2 , i.e. values $+3$, $+2$, $+1$, and $+0$ appear at V_1 and values $+1$ and $+0$ appear at V_2 . Since the output of the second local quantizer (Q_2) is differentiated and added to the output of the first local quantizer (Q_1), output Y yields 11 different values. A high output level and wide dynamic range are thus achieved by quantizing the 16 bit, 4fs data from the digital filter at a sampling frequency 8 times as high (32fs).

PWM Block

The PWM block performs Pulse-Width Modulation on the 11 value digital data from the MASH block to convert into analog signals with 11 different pulse widths which are precision controlled by a crystal controlled oscillator.

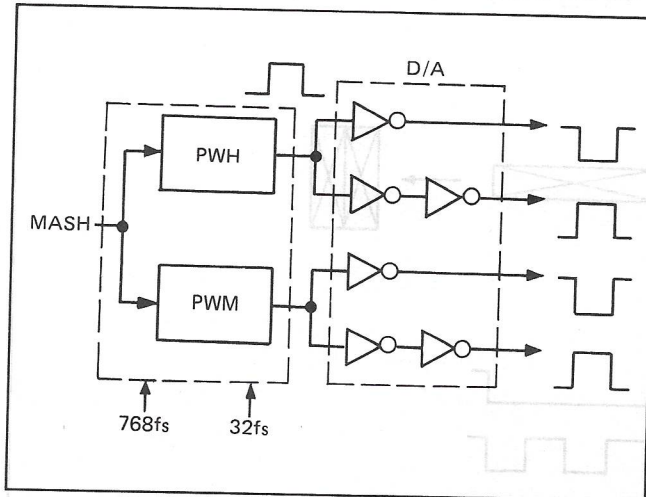


Figure 19-17 PWM Block Diagram

With the PAM, PWM, and PPM techniques, the signal is sampled along the time axis but is not quantized along the amplitude axis. All of these techniques preserve the amplitude information exactly as it was the original analog signal. (see the following figure 19-18)

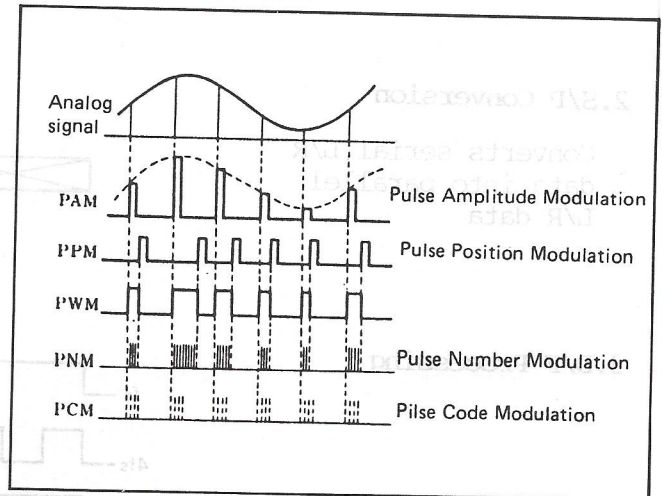
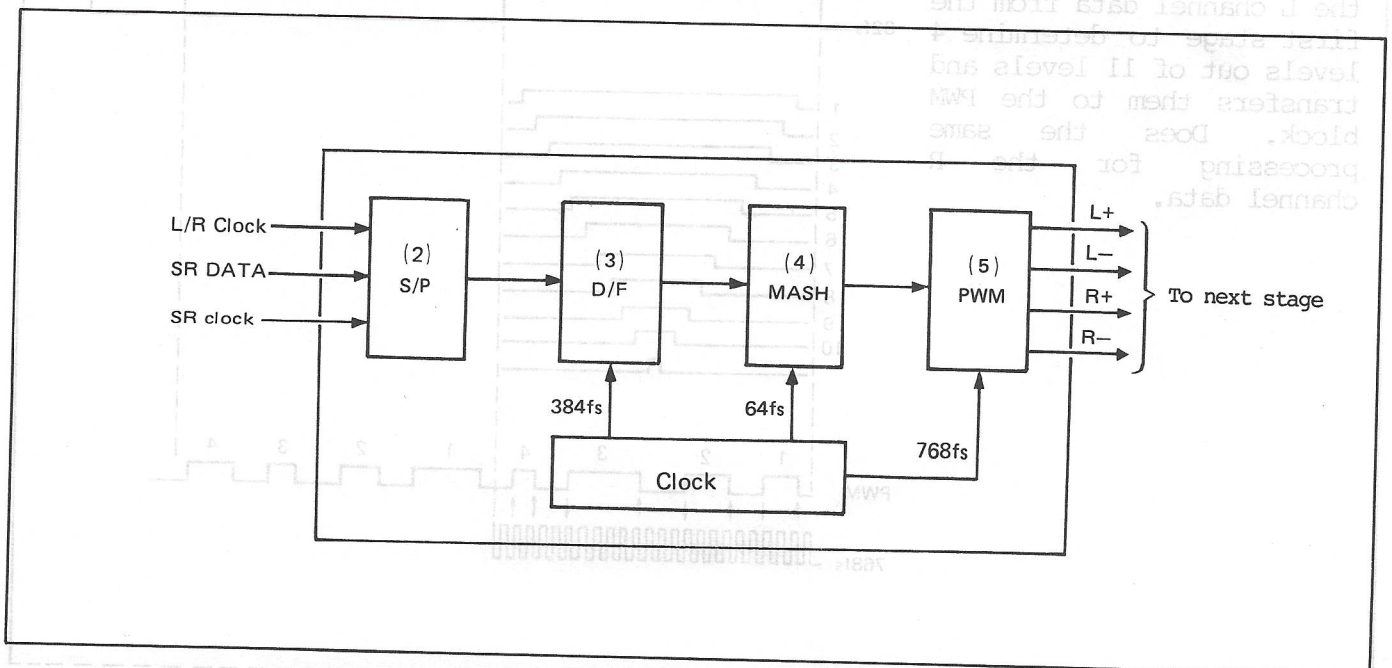


Figure 19-18

There are several different pulse modulation techniques available to convert analog signals into pulse signals.

The pulse width modulation (PWM) samples the input signal along the time axis and yields pulses with pulse widths that correspond to sampled signal amplitudes. The resulting pulse signal contains frequency components similar to those obtained with the PAM technique.

Overall MN6623 Operations



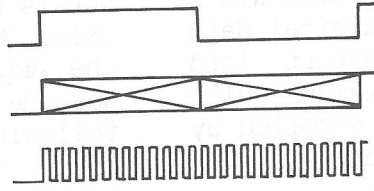
MN6623 Block Diagram

1. Input Signal

L/R Clock

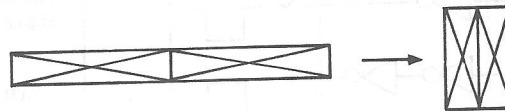
SR Data

SR Clock



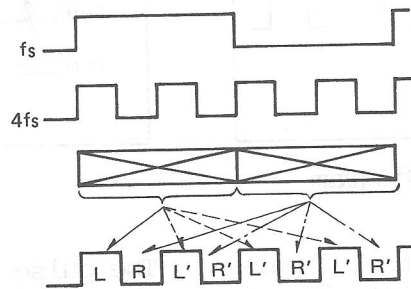
2. S/P Conversion

Converts serial L/R data into parallel L/R data



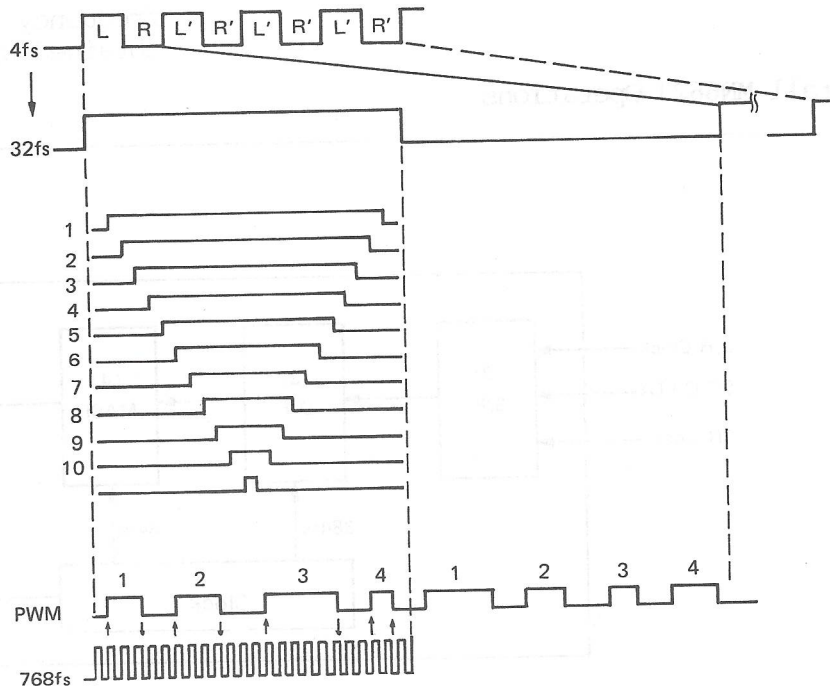
3. D/F Processing

4-hold oversampling



4. MASH Processing

Performs MASH processing on the L channel data from the first stage to determine 4 levels out of 11 levels and transfers them to the PWM block. Does the same processing for the R channel data.



20. Analog Circuit

The MA6623 (IC-801) uses independent +5 volts power supplies for its digital and analog processing block. Either channel of the analog audio signals (+) and (-) at the output of the D/A converter have the signal formats as shows in figure 20-1.

The positive and negative output of the D/A converter are coupled to the differential input of the comparator (IC-803), where they are amplified into a single analog signal with an amplitude twice as large. The output of the comparator is then waveform shaped by the following ternary LPF (IC-805) into a complete analog audio signal.

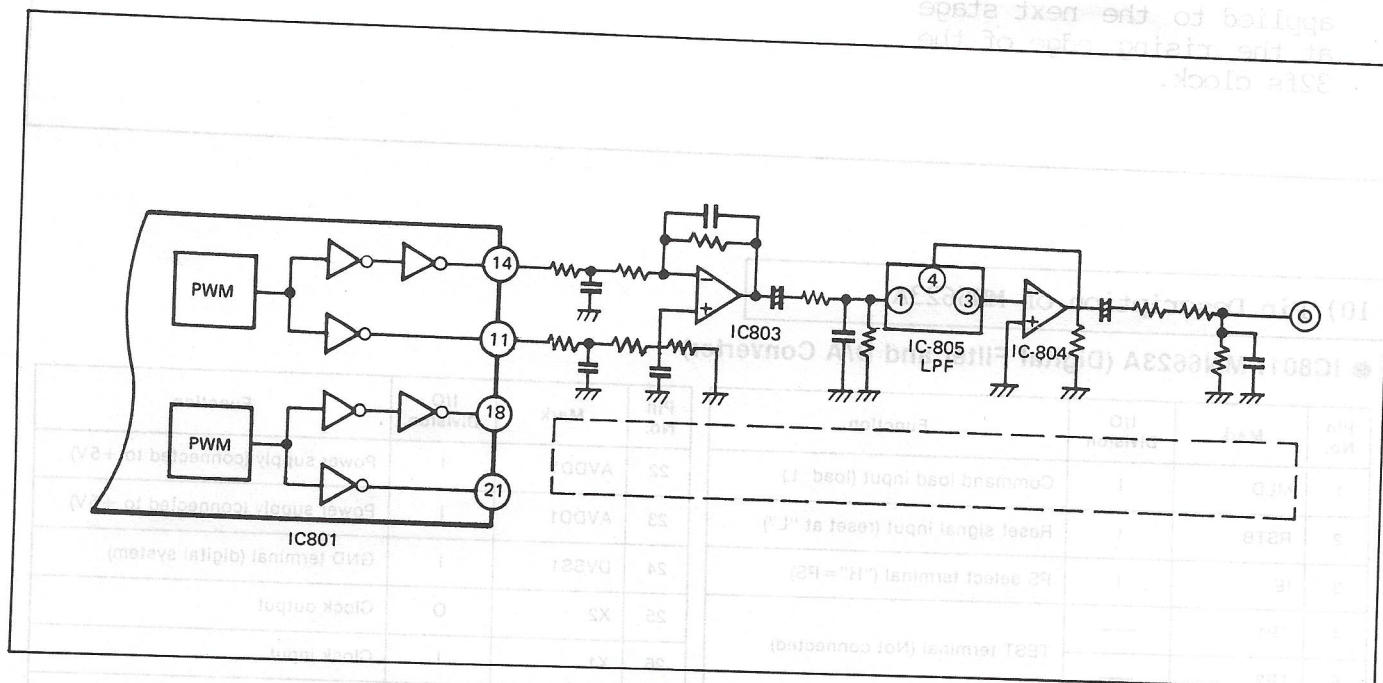


Figure 20-1

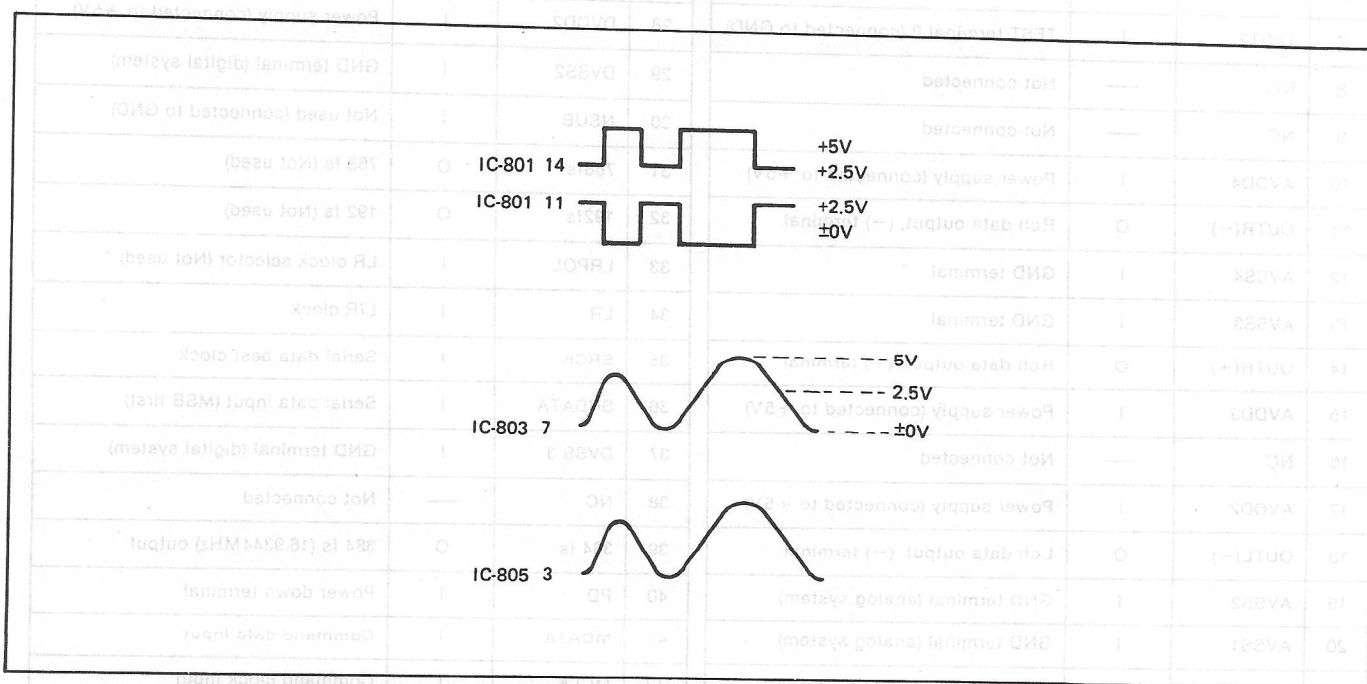
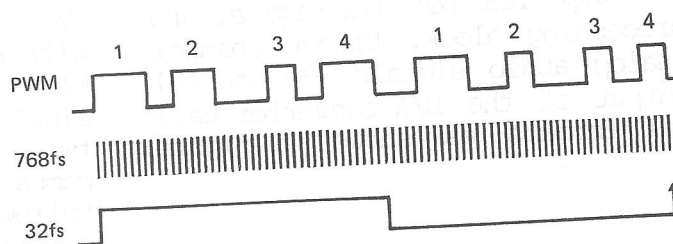


Figure 20-2

5. PWM Processing

Pulse Width Modulates the 4 level data from the MASH block at a rate of 768fs, then separates the L and R channel data with 32fs clock. The L and R channel outputs of the PWM have the same phase, and are applied to the next stage at the rising edge of the 32fs clock.



10) Pin Description of MN6623A

● IC801: MN6623A (Digital Filter and D/A Converter)

Pin No.	Mark	I/O Division	Function
1	MLD	I	Command load input (load: L)
2	RSTB	I	Reset signal input (reset at "L")
3	IE	I	I ² S select terminal ("H"=I ² S)
4	TP1	—	TEST terminal (Not connected)
5	TP2	—	
6	TEST1	I	TEST terminal 1 (connected to GND)
7	TEST2	I	TEST terminal 2 (connected to GND)
8	NC	—	Not connected
9	NC	—	Not connected
10	AVDD4	I	Power supply (connected to +5V)
11	OUTR(-)	O	Rch data output, (-) terminal
12	AVSS4	I	GND terminal
13	AVSS3	I	GND terminal
14	OUTR(+)	O	Rch data output, (+) terminal
15	AVDD3	I	Power supply (connected to +5V)
16	NC	—	Not connected
17	AVDD2	I	Power supply (connected to +5V)
18	OUTL(-)	O	Lch data output, (-) terminal
19	AVSS2	I	GND terminal (analog system)
20	AVSS1	I	GND terminal (analog system)
21	OUTL(+)	O	Lch data output, (+) terminal

Pin No.	Mark	I/O Division	Function
22	AVDD1	I	Power supply (connected to +5V)
23	AVDD1	I	Power supply (connected to +5V)
24	DVSS1	I	GND terminal (digital system)
25	X2	O	Clock output
26	X1	I	Clock input
27	NC	—	Not connected
28	DVDD2	I	Power supply (connected to +5V)
29	DVSS2	I	GND terminal (digital system)
30	NSUB	I	Not used (connected to GND)
31	768fs	O	768 fs (Not used)
32	192fs	O	192 fs (Not used)
33	LRPOL	I	LR clock selector (Not used)
34	LR	I	L/R clock
35	SRCK	I	Serial data best clock
36	SRDATA	I	Serial data input (MSB first)
37	DVSS 3	I	GND terminal (digital system)
38	NC	—	Not connected
39	384 fs	O	384 fs (16.9344 MHz) output
40	PD	I	Power down terminal
41	MDATA	I	Command data input
42	MCLK	I	Command clock input

21. CD Changer's Operation

This section explains the SL-P405C CD changer's mechanism operation's and mechanism logic control operations (including SL-P400C etc).

1. General

The SL-P405C incorporates a recently-developed disc loading unit with an elevating mechanism which enables the user to load up to six compact discs in a magazine and play them in the desired sequence. The SL-P405C uses the One Beam system (FF-1MK2) for its optical (laser) pick-up.

The disc magazine allows loading and unloading of individual discs. The disc detection mechanism (provided only on selected models) provides disc search and displays the presence of discs on the player's display.

The mechanism section uses three DC motors for: Disc Loading, Elevator, and the Spindle. These motors are controlled by the system CPU (IC-401).

The SL-P405C signal processing logic and audio circuit are almost identical to those used in our comparable CD players.

2. Mechanism Orientation

The mechanism can be divided in three basic sections: The magazine holder, the Disc Loader, and the Optical Deck. See figure 21-1.

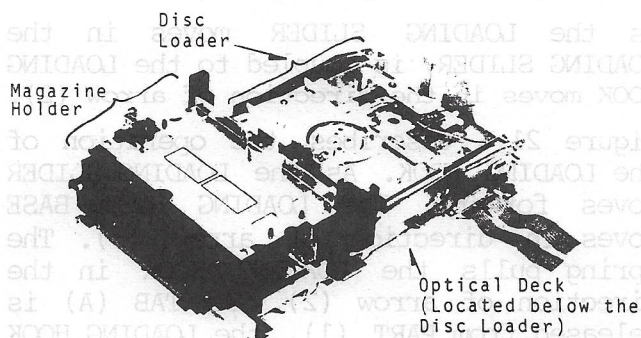


Figure 21-1 Mechanism

- * The magazine holder is located at the front-end of the mechanism. It holds the magazine in place, containing six discs (on trays), ready to be accessed by the disc holder.
- * The disc loader is located behind the magazine holder. Its function is to access the selected disc tray from the magazine and lower it on the optical deck's turntable.
- * The optical deck is located below the disc loader. Its function is to control the optics used to read the disc's data for Audio signal reproduction. The optical deck is the heart of the CD player. Its operation is explained on page 31.

2-1. Costruction of Disc Magazine

Figure 21-1 shows the construction of the disc magazine. The magazine can store up to six disc at a time.

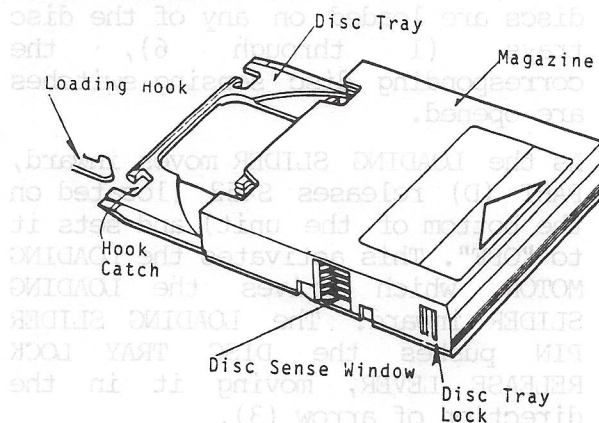


Figure 21-2 Disc Magazine

- (1). Hook Catch:
The disc loader accesses a tray by latching onto the HOOK CATCH of the disc tray.
- (2). Disc Sense Window:
This window is provided so that the six disc sensing switches (provided only on selected models) detect the presence or absence of a disc.
- (3). Disc Tray Lock:
The DISC TRAY LOCK retains the DISC TRAYS in place so that they will not drop out of the magazine accidentally. When replacing a disc, the user pushes this lock while pulling the DISC TRAY.

2-2. Construction and Operation of the Magazine Holder

A disc tray pulled out or returned to the magazine by the LOADING SLIDER which is driven by the LOADING MOTOR via the LOADING GEAR.

The tray to be transferred may be unlocked only at the moment when the tray is caught by the LOADING HOOK and pulled.

- (1). The magazine is inserted into the magazine holder and pushed in the direction of arrow (1). see figure 21-3.
- (2). The magazine first comes in contact with TAB (A) of the EJECT LEVER. see figure 21-3. At the same time PART (B) of the EJECT LEVER comes in contact with the LOADING SLIDER PIN and pushes the LOADING SLIDER in the direction of arrow (2). shows figure 21-4. PART (C) of the EJECT LEVER pushes LEVER (A) which causes the DISC SENSING SWITCH ASSEMBLY to move in the direction of arrow (3). When discs are loaded on any of the disc trays (1 through 6), the corresponding disc sensing switches are opened.
- (3). As the LOADING SLIDER moves inward, PART (D) releases S452 (located on the bottom of the unit) and sets it to "OFF". This activates the LOADING MOTOR, which drives the LOADING SLIDER inward. The LOADING SLIDER PIN pushes the DISC TRAY LOCK RELEASE LEVER, moving it in the direction of arrow (3).

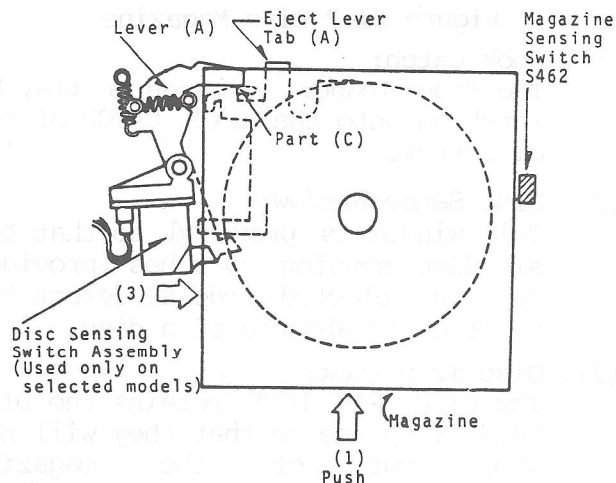


Figure 21-3

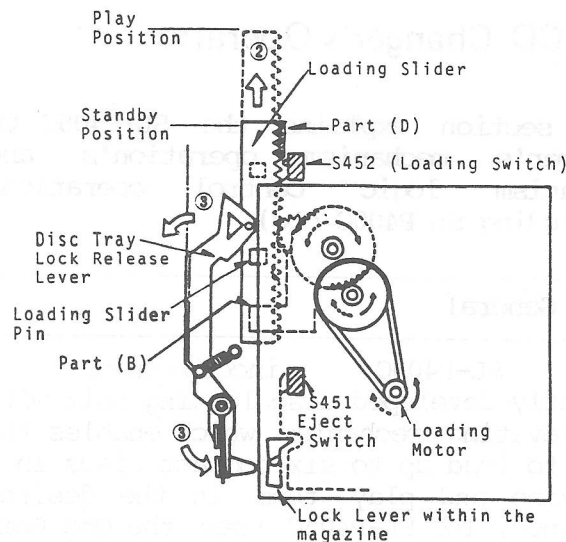


Figure 21-4

- (4). On the right side of the magazine holder, you find the MAGAZINE SENSING SWITCH which is closed when the magazine is fully loaded.

Figure 21-4 shows the disc tray release mechanism within the magazine. The disc trays are locked so that they will not drop out of the magazine during transit. The trays must be unlocked before one of them can be transferred to the player's optical deck.

2-3. Construction and Operation of the Disc Loader

When selecting a disc, the LOADING MECHANISM must access the disc tray from the magazine and transfer it above the optical deck.

The CONNECTION ARM, part of the LOADING SLIDER, is coupled to the LOADING HOOK assembly. shows figure 21-5.

As the LOADING SLIDER moves in the LOADING SLIDER, is coupled to the LOADING HOOK moves in the direction of arrow (2).

Figure 21-6 describes the operation of the LOADING HOOK. As the LOADING SLIDER moves forward, The LOADING HOOK BASE moves in direction of arrow (3). The spring pulls the LOADING HOOK in the direction of arrow (2). As TAB (A) is released from PART (1), the LOADING HOOK then latches to the disc tray catch.

3. Control Logic

3-1. Magazine Loading Operation

(1). The SL-P405C mechanism is equipped with five sensing switches and a height sensor. Four of the sensing switches along with the height sensor. Four of the sensing switches along with the height sensor contribute to the disc magazine loading operation shows figure 21-9. Their purpose is described as follows.

- A. S462 Magazine Detection Switch:
Normally open. Alerts the CPU that a magazine is being loaded.
- B. S451 Eject Switch:
Normally closed. Stops the loading motor operation and senses the "Eject" or (outermost) position of the magazine.
- C. S452 Loading Switch:
Normally closed. Detects the completion of the magazine insertion and activates the loading motor.
- D. S453 Draw Switch:
Normally open. Detects the disc tray's innermost position.

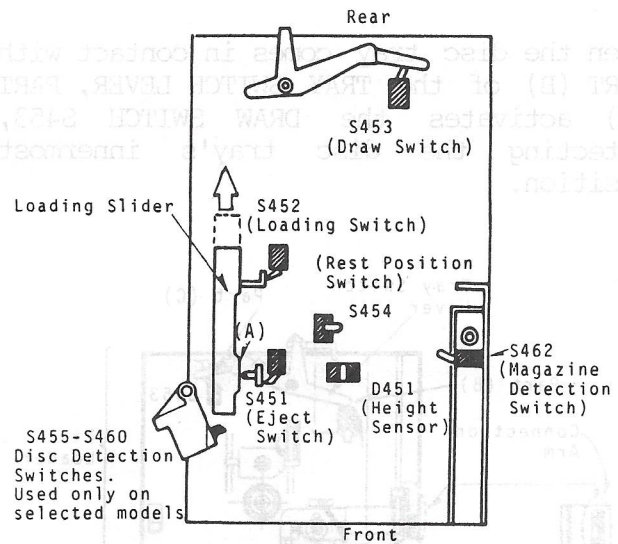


Figure 21-9 Sensing switches location

- E. S454 Rest Position:
Normally open. Detects the elevator unit's lowest position.
- F. D451 Height Sensor:
Detects the elevator's position.

The sensing switches are interfaced with the CPU (IC-401) input, pin 15. shows figure 21-10.

The CPU controls the loading motor and the elevator motor via its output pins 8 through 11, connected to the motor drive circuit of IC-401.

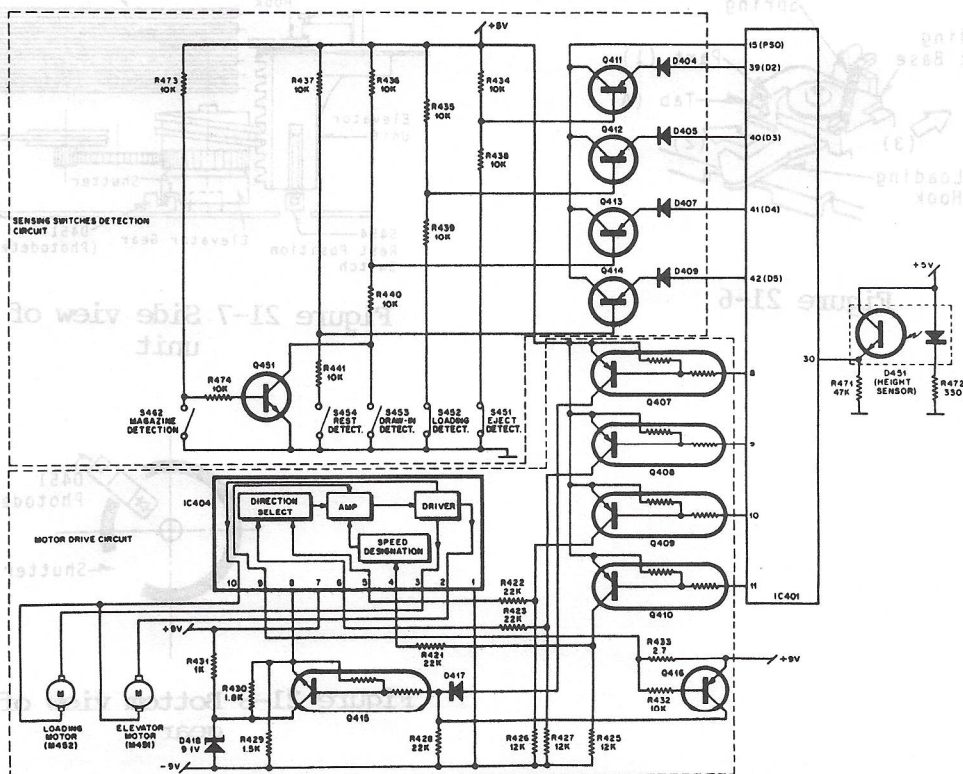


Figure 21-10 Switch sensing and motor drive circuit interface

When the disc tray comes in contact with PART (B) of the TRAY SWITCH LEVER, PART (C) activates the DRAW SWITCH S453, detecting the disc tray's innermost position.

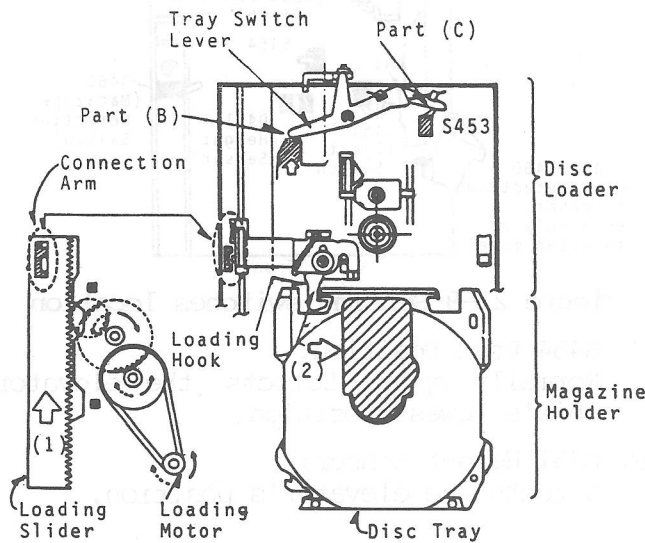


Figure 21-5

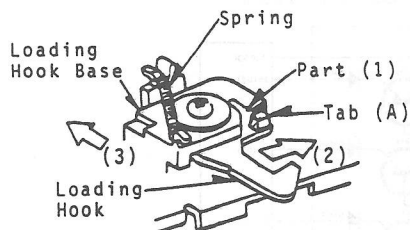


Figure 21-6

2-4. Disc Loader Elevation Mechanism

The function of the elevation mechanism is to drive the disc loader to the selected disc tray.

After a disc has been specified, the ELEVATOR MOTOR drives the ELEVATOR GEAR so that the LOADING UNIT is set at the height of the specified disc. shows figure 21-7.

The bottom of the ELEVATOR GEAR has a SHUTTER with two openings which operate the optical sensing circuit shows figure 21-8.

Two pulses are generated for each rotation of the ELEVATOR GEAR. The DISC LOADER is elevated from one disc to the next each time the ELEVATOR GEAR makes a single revolution.

The purpose of this sensing system is so that the system CPU can determine the location of the ELEVATOR UNIT.

A REST POSITION switch (S454) is provided to sense the lowest position of the ELEVATOR UNIT. This position corresponds to the position at which the disc rests on the turntable.

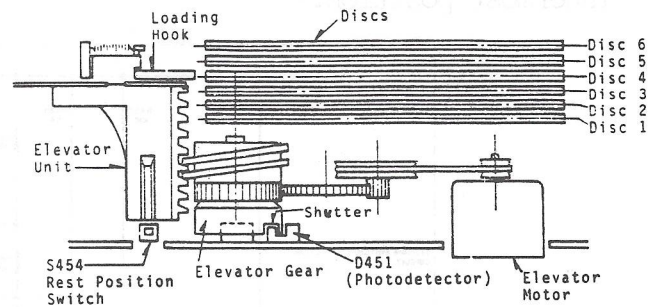


Figure 21-7 Side view of the elevator unit

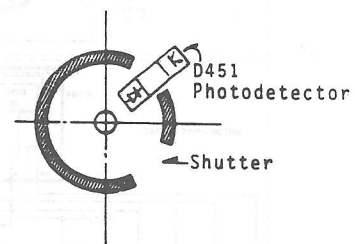


Figure 21-8 Bottom view of the elevator gear

- (2). As the magazine is loaded in its compartment, S462 is first activated alerting the CPU that the magazine is being loaded.

As it is further inserted, the LOADING SLIDER bar first activates S451 and then S452. This causes the loading motor to rotate clockwise and moves the loading slider forward. As the loading slider bar moves forward, PART (A) of the bar comes in contact with S451.

After a pause of approximately 0.24 sec. (Determined by the CPU, the CPU sets pins 8 through 11 so that the elevator motor is selected and rotates in a counterclockwise direction, moving the elevator unit downward).

The elevator unit hits S454 and sets it to the closed position, causing the elevator motor to stop shows figure 21-11.

After a second pause of approximately 0.24 sec, the CPU sets pins 8 through 11 so that the elevator motor is selected and rotates in a clockwise direction. The elevator unit moves fast in the upward direction for approximately 0.16 sec. and then slowly for about 0.08 sec, until the "standby" position (disc 1) is reached.

The above sequence completes the operation from magazine insertion to "standby" position.

MOTOR	MODE	MOTOR DRIVE IC OUTPUT (IC404)			MICROPROCESSOR OUTPUT (IC401)			
		2	3	10	8	9	10	11
Elevator	Up	L	H	H	L	H	H	L
	Down	H	L	L	L	L	H	L
Loading	Pull-out	L	H	L	H	L	L	H
	Return	H	L	H	L	H	L	H

Table 1 Truth table of microprocessor - motor drive circuit

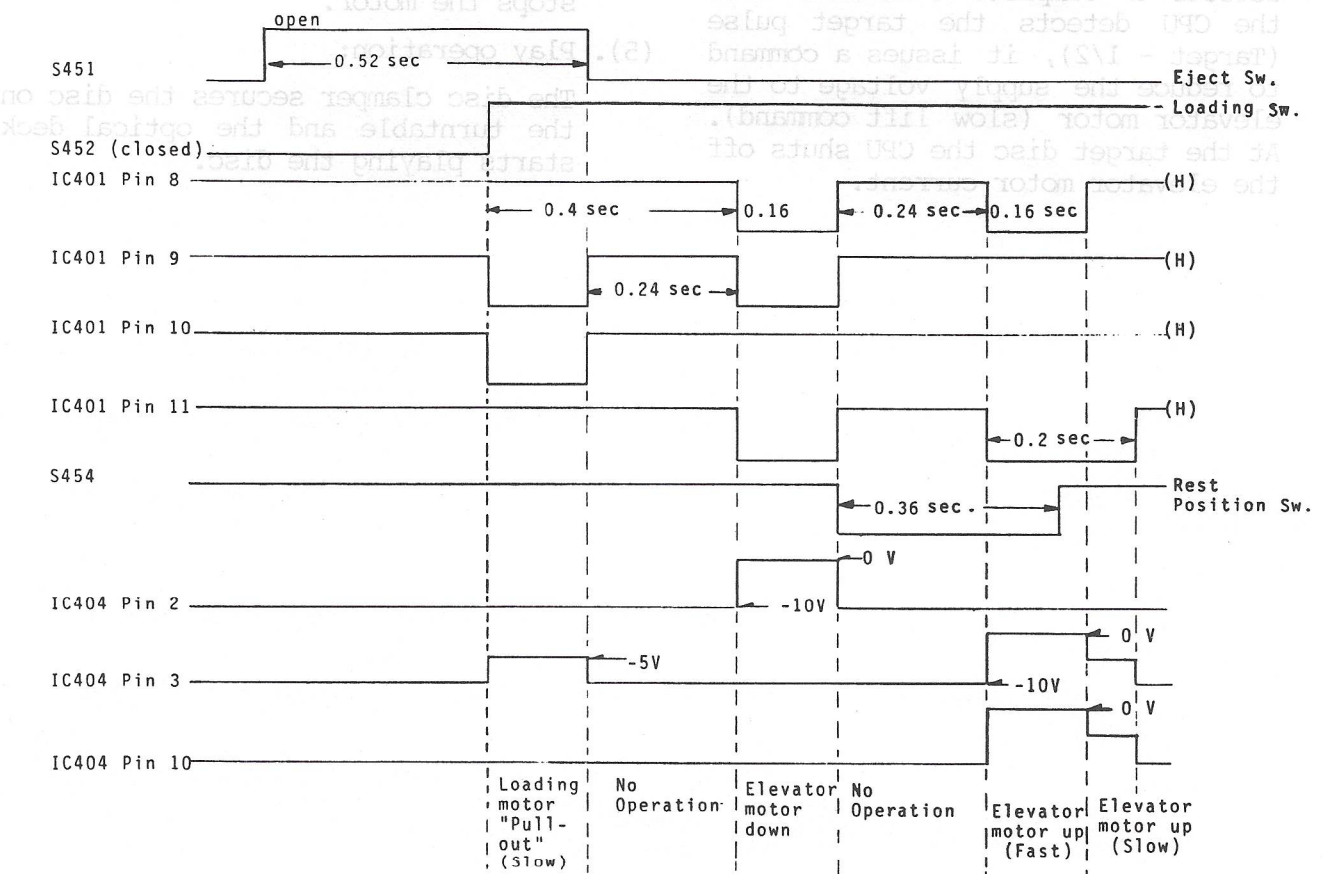


Figure 21-11 Mechanism operation sequence from magazine loading to "standby"

3-2. Disc Loading Operation

After the magazine has been loaded, the elevator unit remains in the "standby" mode, until a disc is selected. The following example will describe the operation sequence to select the third disc and play it shows figure 21-12.

(1). Magazine loading to standby operation:

The magazine is inserted and the unit enters the "standby" mode as described in the previous section.

(2). Disc loading operation:

When "Disc 3" is selected, the CPU (IC-401) energizes the elevation motor to lift the disc loader. At this time the elevator gear generates two pulses per rotation.

The CPU monitors the rising edge of the height sensor (D451) signal.

The disc loader elevates from one disc to the next each time the CPU detects a complete rotation. When the CPU detects the target pulse (Target - 1/2), it issues a command to reduce the supply voltage to the elevator motor (slow lift command). At the target disc the CPU shuts off the elevator motor current.

(3). Disc tray "pull-out" sequence:

The loading motor is energized to pull out the third disc tray. When the CPU detects that the draw switch (S453) is closed, it reduces the supply voltage to the loading motor. When the CPU subsequently detects that the loading switch (S452) is closed, it shuts off the loading motor current.

(4). Disc clamp operation:

Since the optical deck is located two disc span below the first disc, the loading units must be lowered to the C (clamp) position. To do this, the CPU energizes the elevator motor so that it will lower the disc loader. This time, the CPU counts the falling edge of the height sensor pulses. When the CPU counts 8 pulses, it brakes the elevation motor, and when it detects the falling edge of the 9th pulse, it stops the motor.

(5). Play operation:

The disc clasper secures the disc on the turntable and the optical deck starts playing the disc.

3-3. Stop and Eject Operation

Figure 21-13 shows a timing of transition from play to stop and back to play modes. During the tray return sequence, the loading motor is driven in the reverse direction to release the tray loading hook from the hook catch.

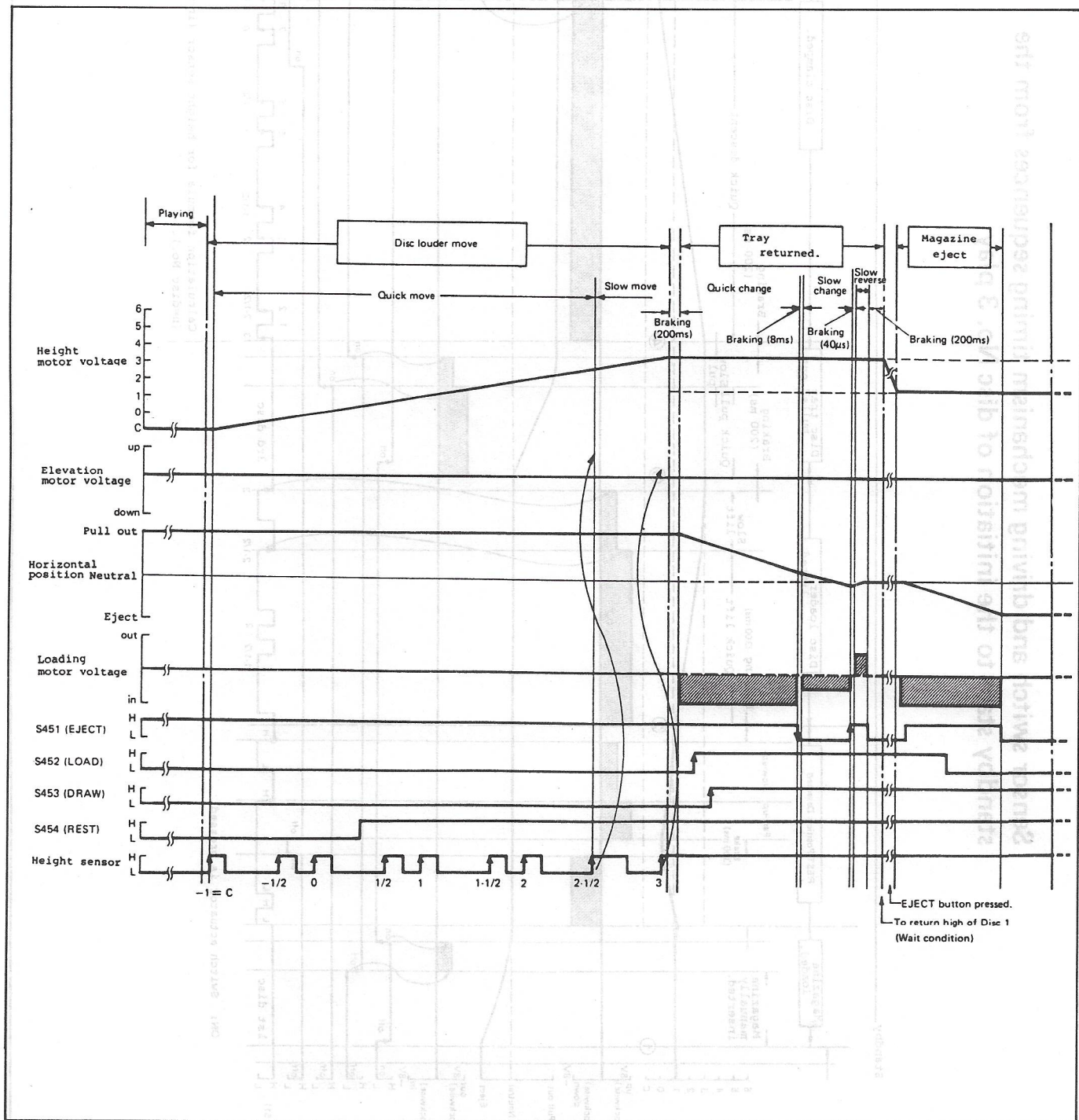


Figure 21-13 Timing chart

Sensor switch and driving mechanism timing sequences from the standby state to the initiation of disc No. 3 play

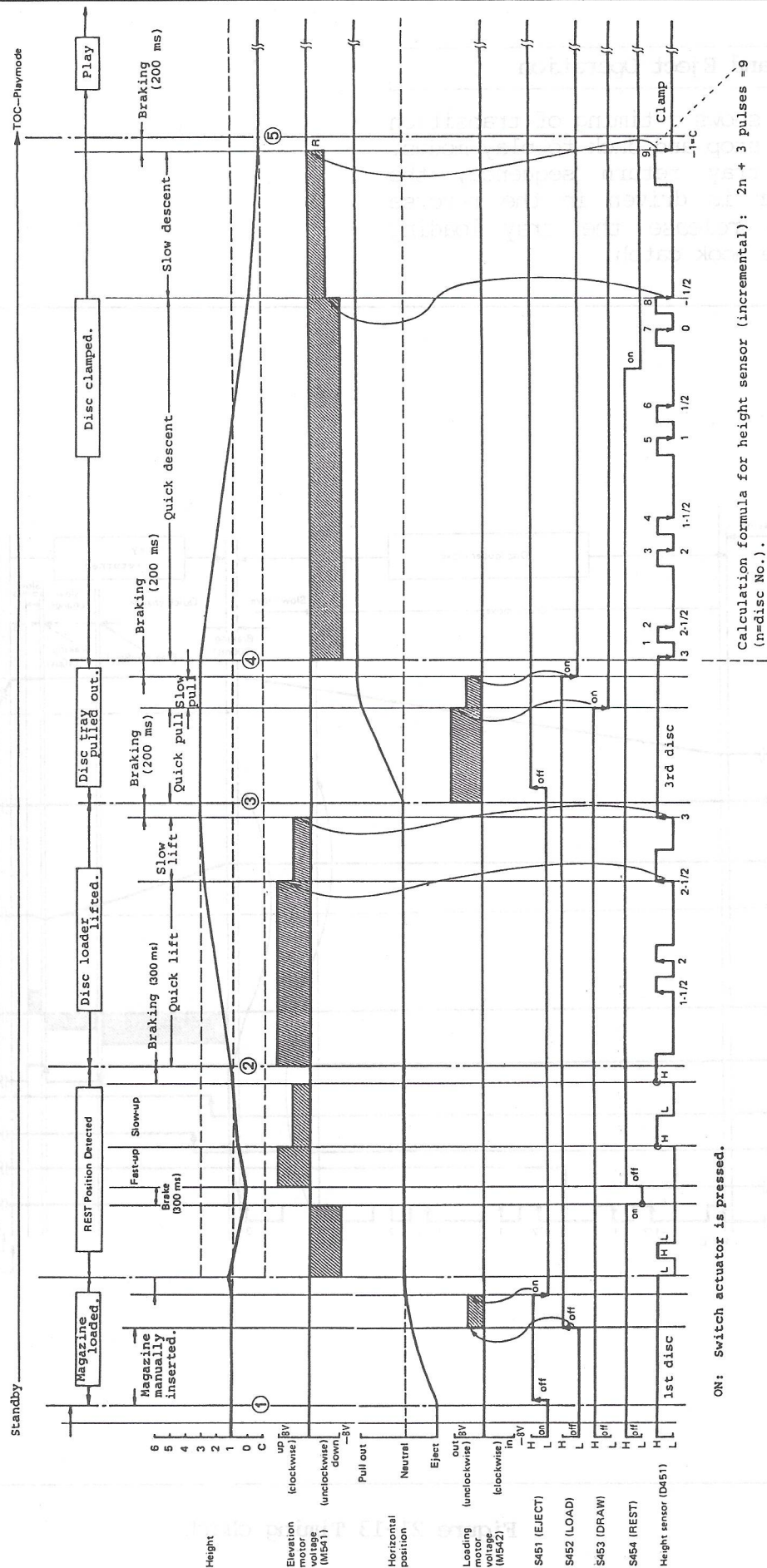


Figure 21-12 Sensing switches and driving motor timing sequence from "standby" to "play" of the third disc